

Supplementary information for: Destruction of long-range order in non-collinear two-dimensional antiferromagnets by random-bond disorder

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(Dated: February 9, 2018)

I. SINGLE BOND DEFECT

Here we provide a few details concerning the physics of a single defect bond.

A. Spin texture via linear response

As explained in the main text, the spin texture induced by a bond defect can be understood as the response to a local dipolar transverse field which acts to rotate the two spins on the defect bond away from their clean-limit configuration. This is most efficiently calculated in a local frame where the clean-limit state is uniform.

To this end we start from a coplanar state in the $x-y$ plane and rotate the spins such that all point along the $\hat{x}$ axis. The rotation due to the dipolar field involves only the y components of the spins, i.e., the susceptibility $\chi^\perp(\vec{q}) = \langle S_{\vec{q}}^y; S_{-\vec{q}}^y \rangle$. For a system with underlying SU(2) spin symmetry, this susceptibility is governed by the in-plane Goldstone modes and generically behaves as $\chi^\perp(\vec{q}) \propto 1/q^2$ for small momenta. For the triangular-lattice Heisenberg model in the classical limit, the full expression for the transverse susceptibility is given by [1]

$$\chi^\perp(\vec{q}) = S \frac{1}{\gamma(\vec{q}) - 1} \quad (\text{S1})$$

where the triangular lattice structure factor

$$\begin{aligned} \gamma(\vec{q}) &= \frac{1}{6} \sum_{\vec{\delta} \in \text{NN}} e^{i\vec{q} \cdot \vec{\delta}} \\ &= \frac{1}{3} \left(\cos q_x + 2 \cos \frac{q_x}{2} \cos \frac{\sqrt{3}}{2} q_y \right) \end{aligned} \quad (\text{S2})$$

encodes the geometry of the lattice. For a bond defect at a position $\vec{r}'$ on a coupling along the direction $\vec{\delta}(\vec{r}')$, the transverse response at a separate site $\vec{r}$ is given by

$$\langle S_{\vec{r}}^\perp \rangle = \sum_{j=0}^1 h_{\vec{r}'}^\perp \beta_j \langle S_{\vec{r}}^y; S_{\vec{r}'+\beta_j \vec{\delta}(\vec{r}')/2}^y \rangle \quad (\text{S3})$$

where $\beta_j = (-1)^j$ is the dipolar form factor of the perturbation, and $h_{\vec{r}'}^\perp \propto \delta J/J$ measures the strength of the perturbation due to the defect at bond location $\vec{r}'$. The

proportionality factor, which we call $\tilde{\alpha}$, depends on the microscopic details of the lattice and in itself is proportional to the stiffness of the ordered phase. Later, we extract the values of this non-universal prefactor for two different non-collinear antiferromagnets, the triangular-lattice Heisenberg model and the $J_1 - J_4$ cubic-lattice Heisenberg model. At this stage the co-ordinate locations of the bonds and the spins are considered separately. However, when we take the appropriate continuum limit the distinction will disappear.

Once the transverse response is expressed in terms of the susceptibility $\chi^\perp(q)$,

$$\langle S_{\vec{r}}^\perp \rangle = \frac{\tilde{\alpha}}{J} \sum_{\vec{q}} \delta J(\vec{r}') \tilde{\beta}_{\vec{\delta}(\vec{r}')}(\vec{q}) \chi^\perp(\vec{q}) e^{i\vec{q} \cdot (\vec{r} - \vec{r}')} \quad (\text{S4})$$

with the momentum dependence of the form factor given by $\tilde{\beta}_{\vec{\delta}(\vec{r}')}(\vec{q}) = e^{-i\vec{q} \cdot \vec{\delta}(\vec{r}')/2} - e^{i\vec{q} \cdot \vec{\delta}(\vec{r}')/2} = -2i \sin(\vec{q} \cdot \vec{\delta}(\vec{r}')/2)$, its dipolar nature becomes apparent. The long wave length response,

$$\langle S_{\vec{r}}^\perp \rangle = \frac{\alpha S}{J} \sum_{\vec{q}} \delta J(\vec{r}') \frac{\vec{q} \cdot \vec{\delta}(\vec{r}')}{q^2} e^{i\vec{q} \cdot (\vec{r} - \vec{r}')} \quad (\text{S5})$$

and thereby the spin texture of the resulting state is determined by the defect induced dipole moment $\vec{d}(\vec{r}') = (\delta J(\vec{r}')/J) \vec{\delta}(\vec{r}')$ and the quadratic dispersion of the Goldstone modes. Once again the prefactor α (see also in main text) depends on the underlying lattice and following (S2), for triangular lattice, we get $\alpha = 4i\tilde{\alpha}$.

B. Uniform moment

In the presence of a single bond defect, the bulk spin moments do not exactly cancel each other, such that the resulting state has a non-vanishing uniform magnetization m_{imp} . For small δJ this is proportional to δJ ,

$$m_{\text{imp}} = \gamma S \delta J. \quad (\text{S6})$$

For the triangular-lattice Heisenberg model we numerically find $\gamma = 0.30$, see Fig. 1B. Our estimate of γ includes systematic $1/S$ correction to the classical value. The result was obtained through semiclassical spin-wave approximation with mean-field angle correction included.

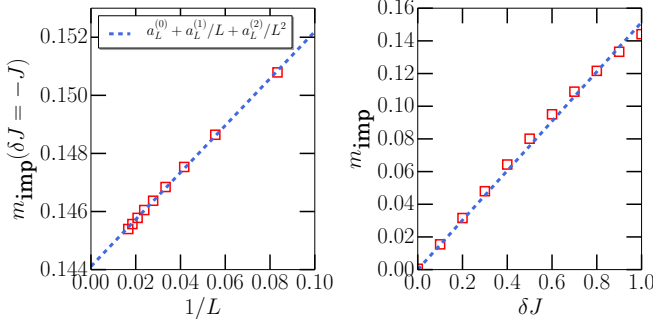


Figure S1. Uniform moment generated due to bond defect δJ . (a) Finite-size scaling of the uniform moment for an absent bond at the center of the lattice. The scaling co-efficients were extracted to be, $a_L^{(0)} = 0.1441$, $a_L^{(1)} = 0.0809$ and $a_L^{(2)} = 1.0$. **[ToDo: This comes out of the fitting but I suppose this is not appropriate? Should we revert to a linear fitting?]** (b) Growth of the uniform moment with increasing defect amplitude δJ . For small values of δJ the relationship is linear.

The proportionality can be expected to break down for larger δJ , but still holds to very good accuracy **[ToDo: There is a problem here. The moment and its 1/S correction are vectors and the sum their magnitude is greater than the magnitude of their sum. For the total γ I sum the vectors and then get their magnitude while for the 1/S expansion I collect the spin-wave correction separately. This needs to be clarified.]** for $\delta J = -J$ where we find $\gamma = 0.39 - 0.08S^{-1} + \mathcal{O}(S^{-2})$.

A finite-size scaling analysis of m_{imp} is shown in Fig. IB.

II. DESTRUCTION OF LONG-RANGE ORDER

As explained in the main text, an estimate of the transverse spin fluctuations induced by a collection of bond defects can be used to assess the destruction of long-range order.

In the linear response regime, we can consider the superposition of individual dipolar responses (S5) from a spatially random distribution of bond defects. The averaged transverse response is simply zero for a defect distribution with zero mean. Therefore their collective effect is manifested through the squared transverse fluctuation. Although the angular profile of the response due to each bond defect is dependent on the orientations of the respective bonds, for the case of a perfectly random distribution of such defects, $\delta J(\vec{r})\delta J(\vec{r}') = (\delta J)^2 \delta_{\vec{r}, \vec{r}'}$, the contribution coming from the different bond or dipolar orientations decouple. From (S5) it is also easy to see that these contributions differ by phase factors of constant angular rotations. Therefore, by choosing appro-

priately rotated frames of axes for each, we can cast them in an identical form and sum up to find the expression for the total transverse spin fluctuation,

$$\begin{aligned}
 \overline{\langle S_0^{\perp 2} \rangle} &= \left(\frac{\alpha}{J} \right)^2 \sum_{\vec{r}, \vec{r}'} \sum_{\vec{q}, \vec{q}'} \overline{\delta J(\vec{r}) \delta J(\vec{r}')} \\
 &\quad (\vec{q} \cdot \vec{\delta}(\vec{r})) (\vec{q}' \cdot \vec{\delta}(\vec{r}')) \chi^\perp(\vec{q}) \chi^\perp(\vec{q}') e^{i(\vec{q} \cdot \vec{r} + \vec{q}' \cdot \vec{r}')} \\
 &= \left(\frac{\alpha \delta J}{J} \right)^2 \sum_{\vec{r}} \sum_{\vec{q}, \vec{q}'} (\vec{q} \cdot \vec{\delta}(\vec{r})) (\vec{q}' \cdot \vec{\delta}(\vec{r})) \\
 &\quad \chi^\perp(\vec{q}) \chi^\perp(\vec{q}') e^{i(\vec{q} + \vec{q}') \cdot \vec{r}} \\
 &= \left(\frac{\alpha \delta J}{J} \right)^2 \frac{1}{N_B} \sum_i \sum_{\vec{q}} (\vec{q} \cdot \vec{\delta}_i) (\vec{q} \cdot \vec{\delta}_i) \chi^\perp(\vec{q}) \chi^\perp(\vec{q}') \\
 &= \left(\frac{\alpha \delta J}{J} \right)^2 \int \frac{d^d q}{(2\pi)^d} (q_x \chi^\perp(\vec{q}))^2
 \end{aligned} \tag{S7}$$

where $\chi^\perp(q) \propto S/q^2$ is the transverse susceptibility from the previous section and N_B is the number of distinct bond orientation for a given lattice geometry. It is noteworthy that the additional factor from the summation over the bond orientations is cancelled by the overall volume fraction for each such contribution. **[ToDo: I think it does, do you?]** As discussed in the foremost part of this letter, if the susceptibility $\chi^\perp(\vec{q})$ is dominated by massless Goldstone modes, the integral (S7) diverges below a lower critical dimension. This signals the destruction of the long-range order and precisely at the critical dimension gives rise to the emergence of exponentially growing domains. In the following we consider the criterion $\overline{\langle S^{\perp 2} \rangle} \simeq S^2$ and provide a detailed supplement to our original discussion on various aspects of the destruction of long-range order in bond-disordered non-collinear magnets.

A. SU(2) symmetry and $T = 0$

For spontaneously broken SU(2) symmetry in a magnetic system the fundamental massless excitations have quadratic dispersions and therefore the transverse susceptibility $\chi^\perp(\vec{q}) \propto 1/q^2$. The convergence criterion of the integral (S7) makes the correlation length of the order parameter short-ranged,

$$\frac{1}{(2\pi)^{d-1}} \left(\frac{\alpha S \delta J}{\sqrt{2} J} \right)^2 \int_{1/\xi}^1 q^{d-1} dq \left(\frac{1}{q} \right)^2 \simeq S^2. \tag{S8}$$

This implies a correlation length $\xi^{2-d} \propto (J/\delta J)^2$ for $d < 2$, while for $d = 2$, barring the numerical and non-universal prefactors,

$$\xi \propto e^{(J/\delta J)^2}. \tag{S9}$$

[ToDo: Should I be more precise?]

B. Finite T

At finite T and in [ToDo: in or at?] $d = 2$ the magnet would be thermally disordered, but similar to the effect of bond disorder from the previous segment it would exhibit exponential correlation growth, $\xi \propto e^{1/T}$ [2]. As temperature decreases, the thermal and quenched bond disorder competes for precedence and for a fixed value of disorder strength a crossover temperature emerges, $T^* = \delta J/J$. If the temperature decreases any further the disordered state becomes magnetic. [ToDo: This is apparently contentious now, so we should change may be? In Chakravorty Halperin, I see a factor of spin-stiffness in the exponent, should we try to be precise and try get an appropriate cancellation of this with our prefactor α ? Although, don't you think comparing our bare values to the renormalized stiffness from Chakravarty et. al would be comparing apples to oranges?]

C. Layered systems

A realistic layered system displays a small inter-layer coupling which tends to stabilize magnetic order at finite temperatures in the presence of SU(2) symmetry. We assume an (unfrustrated) inter-layer coupling of magnitude $J_\perp$. Using the ratio between the interplanar and intraplanar couplings, $\kappa = J_\perp/J$ we can express the long-wavelength transverse susceptibility of the coupled layered system as

$$\chi^\perp(\vec{q}, q_\perp) = \frac{\lambda S}{q^2 + \kappa^2 q_\perp^2} \quad (\text{S10})$$

where as before we have a dimensionless lattice dependent factor λ . Substituting this expression in the integral relation (S7) it becomes apparent that only the impurities among the in-plane bonds affect the long-range order. The stability criterion is determined by the coupling ratio κ alone,

$$\left(\frac{\lambda \alpha S \delta J}{J}\right)^2 \int \frac{d^2 q d q_\perp}{(2\pi)^3} \left(\frac{q_x}{q^2 + \kappa^2 q_\perp^2}\right)^2 \leq S^2 \quad (\text{S11})$$

with the long-range ordered phase stable if,

$$\frac{J_\perp}{J} = \kappa \geq \frac{1}{16\pi} \left(\frac{\lambda \alpha \delta J}{J}\right)^2 \quad (\text{S12})$$

D. Broken SU(2)

For a system with broken SU(2) symmetry, some or all of the Goldstone modes acquire a gap. For the latter

situation, we assume a generic form of the transverse response function

$$\chi^\perp(\vec{q}) = \frac{\lambda S}{q^2 + \Delta^2} \quad (\text{S13})$$

with the dimensionless phenomenological factor λ absorbing the particular details of the lattice. The gap removes the infrared divergence of the integral (S7) and the stability criterion for the long-range order is solely determined by the amplitude prefactor of the integral. With the presently adopted parametrization, the critical minimum value of the gap which stabilizes the long-range order is

$$\Delta^{2-d} \geq \frac{1}{(2\pi)^{d-1}} \left(\frac{\lambda \alpha S \delta J}{\sqrt{2} J}\right)^2. \quad (\text{S14})$$

Below this value the correlation length decays as a power law of the gap magnitude with $\Delta \rightarrow 0$. [ToDo: I think so, Do you?]

III. INCOMMENSURABILITY IN ANISOTROPIC HEISENBERG MODEL

We consider a spatially anisotropic Heisenberg model on the triangular lattice, with couplings $J_{1,2,3}$ on the bonds 1, 2, 3, with mean J . In the classical limit, the ground state is given by a single- Q coplanar spiral whose propagation wavevector $\vec{Q}$ can be found by minimizing the Fourier-transformed coupling constant $J(\vec{q})$. In the isotropic case, the system orders at $\vec{Q}_0 = \pm(4\pi/3, 0)$.

The degree of anisotropy can be parameterized by $\vec{d} = \sum_{i=1}^3 ((J_i - J)/J) \hat{e}_i$ [ToDo: Note: change of definition, do you agree?] where $\hat{e}_i$ are the directed unit vectors along the nearest-neighbor bonds of the lattice. For the following, we chose the convention, $\hat{e}_1 = (1, 0)$, $\hat{e}_2 = (-1/2, \sqrt{3}/2)$ and $\hat{e}_3 = (-1/2, -\sqrt{3}/2)$. As discussed in the main text and also in the preceeding sections, for localized anisotropies, $\vec{d}$ takes the role of a local dipole density which becomes crucial in the bond-disordered case.

Within the parameter regime $J_1^2(J_2 - J_3)^2 < J_2^2 J_3^2 < J_1^2(J_2 + J_3)^2$, the ordering vector for the anisotropic triangular lattice is given by

$$\begin{aligned} \vec{Q} &= (Q_x, Q_y) \\ Q_x &= 2 \tan^{-1} \left(\frac{-\sqrt{J_1^2(J_2 + J_3)^2 - J_2^2 J_3^2}}{\sqrt{-J_1^2(J_2 - J_3)^2 + J_2^2 J_3^2}} \right) \\ Q_y &= -\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{J_2 - J_3}{J_2 + J_3} \frac{\sqrt{J_1^2(J_2 + J_3)^2 - J_2^2 J_3^2}}{\sqrt{-J_1^2(J_2 - J_3)^2 + J_2^2 J_3^2}} \right) \end{aligned} \quad (\text{S15})$$

For small deviations from isotropy, $|\vec{d}| \ll 1$, the relation between $\vec{Q}$ and $\vec{d}$ gets simpler, $\vec{Q} - \vec{Q}_0 = -\frac{2}{\sqrt{3}} \vec{d}$. However, the most general relation between the shift of the

ordering wave vector and the anisotropy induced dipole moment in an arbitrary lattice with non-collinear order reads,

$$Q^\alpha - Q_0^\alpha = \sum_{i,\alpha,\beta} \omega_i^{\alpha\beta} d_i^\beta \quad (\text{S16})$$

where we sum over dipole moments along different directed bond orientations and their components. **[ToDo: Do we give ω a name? or do we leave this generalization all together?]** It is evident from the above expression that for even small bond anisotropies the ordering wave vector becomes incommensurate.

We can extend this result for the fixed lattice anisotropies to the case of isolated bond defects by noticing that now instead of a single energy functional $\propto J(\vec{q})$, we have to simultaneously minimize multiple distinct local energy functionals, $J_{\vec{r}}(\vec{Q}) = \sum_i 2J_{\vec{r},i} \cos(\vec{Q} \cdot \vec{e}_i)$ at all the sites $\vec{r}$. However, due to the nearest neighbour nature of the exchange interaction, the effect of an isolated infinitesimal defect remains limited to the generation of a local infinitesimal dipole moment $\vec{d}$ on the same location. A random distribution of such bond defects thus lead to rigid incommensurability domains. In the main text we have identified the consequence of the emergence of these domains as the destruction of long-range order.

IV. NUMERICAL SIMULATIONS

[ToDo: Changed it from NUMERICAL RESULTS. Would this be a more proper? Or did you have something else in your mind]

From the discussion in the previous section it is clear that the introduction of bond impurities change the ordering wave vectors for non-collinear magnets. The loss of translational invariance implies that the classical ground state can no longer be analytically obtained for arbitrarily strong bond defects. Therefore, in order to find the consequent spin-texture, dipolar decay and impurity moments **[ToDo: should we call this defect moment?]**, we adopt an iterative classical energy minimization scheme [3] to reach the ground state configuration. For the triangular lattice, we start from the 120° initial configuration and the spins rotate in response to the emergent dipolar moment generated by the bond defect.

[ToDo: Should we discuss a bit about random bond configurations and incommensurability domains here?]

V. BOND DEFECT IN THREE DIMENSIONS

We have checked that our scenario for bond disorder applies to three space dimensions as well. To this end, we

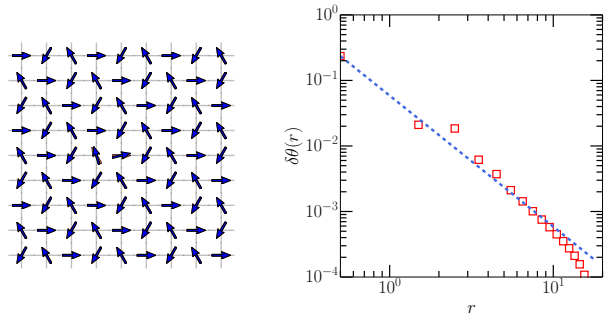


Figure S2. Single bond defect in a $J_1 - J_4$ cubic-lattice Heisenberg antiferromagnet, with an absent bond $\delta J = -J$ (a) Spin-configuration around the absent bond; the unperturbed $(2\pi/3, 2\pi/3)$ order is shown in red. (b) Defect-induced rotation angles $\delta\Theta$ of the spin as function of distance r from the defect. The blue line corresponds to a $1/r^2$ behavior;

have considered a single bond defect in the $J_1 - J_4$ cubic-lattice Heisenberg model. **[ToDo: Should I write a hamiltonian?]** The presence of the J_4 , or the fourth nearest neighbour interaction term in the hamiltonian makes the original cubic-lattice weakly frustrated as the spins can no longer anti-align in pairs to minimize their energies. Ignoring the bonds J_2 and J_3 does not qualitatively affect the discussion that follows as they do not frustrate the spin configuration and in the absence of the J_4 term the ground state remains collinear. **[ToDo: A reference here?]** A finite J_4 , however, renders the ground state spin configuration non-collinear and in general, also incommensurate. The ordering wave vector can be extracted following the same procedure which was applied to the triangular-lattice Heisenberg model. For $J_1 < 4J_4$, we consider the ordering vector,

$$\vec{Q} = (Q, Q, Q) \quad (\text{S17})$$

$$Q = \pi - \tan^{-1} \left(\sqrt{16 \left(\frac{J_4}{J_1} \right)^2 - 1} \right).$$

which is one of the four equivalent ordering vectors of the model in this parameter range.

We take $J_4 = J_1/2$ for which the order is commensurate and non-collinear with ordering vector $\vec{Q} = (2\pi/3, 2\pi/3)$. We introduce an isolated bond defect and find that the dipolar nature of the defect is manifested through the reoriented spin-texture around it (Fig. S2). However, the decay of the transverse response (Fig. S2) is quadratic in this case as the underlying lattice is three-dimensional.

The faster decay of the transverse response significantly affects the long-wavelength behavior of this model. Following on the arguments presented throughout this letter we can easily conclude that being above the lower critical dimension, long-range order in $J_1 - J_4$ cubic lattice is unaffected by random distribution of infinitesimal

bond defects.

- [2] S. Chakravorty, B. I. Halperin, and D. R. Nelson Phys. Rev. Lett. **60**, 1057 (1988).
- [3] L. R. Walker, and R. E. Walstedt, Phys. Rev. Lett. **38**, 514 (1977).

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- [1] A. Wollny, PhD Thesis. (2016).